

SnapBurst - Validated Evidence Summary

Status

EXTERNAL OVERVIEW - CURRENT EXECUTION RECORD

This summary includes only current, directly evidenced, clean PASS campaigns from the SnapBurst validation inventory dated 30 July 2026.

SnapBurst is designed to keep unmanned-aircraft commands authorised, bounded and auditable when communications or command authority become uncertain.

Current primary execution evidence

Evidence record	Recorded result	What it establishes	Current boundary
SB-VAL-008 N=10 C2 loss and restoration	7/7 phases PASS 16/16 FreeFlight transitions 16/16 governed hand-backs 0 foreign events 0 lane leakage 0 safety events 0 stale ACK events	Ten independent PX4 SITL lanes completed single-lane, five-lane and all-lane simulated C2-loss and restoration phases through the natural per-lane verdict path.	Single recorded SITL run. Independent governed lanes, not coordinated multi-aircraft mission behaviour.
SB-VAL-004 Four-hour multi-scenario campaign	14,405 seconds 222/222 tests PASS 8 scenario groups 7 full cycles plus 1 time-limited partial cycle	Sustained current-bridge PX4 SITL execution across command governance, FreeFlight, isolation, recovery, hand-back and multi-lane groups from N=1 through N=10.	Single-host sequential execution. It does not represent simultaneous N=10 endurance.
SB-VAL-005 Governed FreeFlight scale ladder	9/9 rungs PASS N=1, N=2, N=3, N=5 and N=10 0 recorded cross-namespace leakage 0 recorded stale artefacts	Governed FreeFlight dispatch reached the correct lane, while REJECT outcomes blocked dispatch across single-, dual- and multi-lane configurations.	Proves governance and routing of FreeFlight commands, not trajectory accuracy or mission completion.
SB-VAL-007 Dual-bridge isolation	7/7 tests PASS Sequential matrix 4 concurrent verdict combinations State continuity Topic attribution	Two PX4 SITL bridge namespaces remained correctly attributed across PASS, REJECT and mixed-verdict cases with no recorded cross-namespace leakage, stale artefact or wrong-bridge attribution.	N=2 isolation only. Dual-bridge recovery and N>2 coordinated interaction are outside this result.
SB-VAL-011 Real PX4 command acknowledgement	1/1 proof PASS SET_MODE dispatched PX4 result: ACCEPTED ACK correlation confirmed Pending ACK match confirmed	The current bridge routed a governed SET_MODE command to PX4 v1.16.0 SITL and received a real correlated COMMAND_ACK.	Single aircraft and one command surface. ARM, DISARM, MOVE and multi-aircraft ACK behaviour are separate proof categories.

Validated public position

- governed command decisions before aircraft-command egress;
- simulated C2-loss entry into bounded FreeFlight;
- governed restoration of external command and hand-back;
- independent aircraft-lane attribution and isolation;
- PX4 and ROS 2 software-in-the-loop integration;
- real correlated PX4 acknowledgement for the tested SET_MODE command;
- structured, run-specific execution evidence.

Outside the current claim boundary

The current record does not establish hardware-in-the-loop validation, physical-flight validation, certified BVLOS operation, detect-and-avoid performance, distributed swarm coordination, coordinator-loss tolerance or unrestricted autonomous mission planning.

The N=10 evidence demonstrates isolated governed aircraft lanes under coordinated test timing. It does not establish aircraft-to-aircraft mission coordination, leadership transfer, task reassignment or autonomous survivor reconfiguration.